ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

POWER ELECTRONICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 3rd year Power Electronics paper 1 · practice a.

Written by Academic Central for practice, following power electronics modules at South African universities (e.g. University of Pretoria EDF 320 Power Electronics: power semiconductor devices, heat sinks, snubbers and gate drives, AC-DC and DC-DC converters). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct method, sketches that show the required features, and answers that differ only through rounding.

QUESTION 1

1.1 Diode: not controlled; on when forward biased, off when the current reverses; e.g. rectifiers and freewheeling ✓✓ (8)
SCR: turned on by a gate pulse, turns off only when its current falls to zero; slow; very high power, e.g. controlled rectifiers, soft starters ✓✓
MOSFET: fully controlled by gate voltage; very fast (hundreds of kHz); low and medium voltage, e.g. robot motor drivers, DC-DC converters ✓✓
IGBT: fully controlled by gate voltage; medium speed (up to about 20 to 50 kHz); high voltage and current, e.g. variable-speed drives, inverters ✓✓

1.2 Turn on: forward voltage across it and a gate current pulse ✓ (3)
Turn off: it cannot be turned off at the gate; it switches off when the current falls below the holding current ✓, which happens naturally when the AC voltage reverses each half-cycle (natural commutation) ✓

1.3.1 $P = I^2R = 10^2 \times 0.05$ ✓ $= 5$ W ✓ (2)

1.3.2 $P_{sw} = E \times f = 0.2 \times 10^{-3} \times 20\,000$ ✓ $= 4$ W ✓ (3)
$P_{total} = 5 + 4 = 9$ W ✓

1.4 $\theta_{total} = (T_j - T_a) / P = (150 - 40) / 9$ ✓ $= 12.22$ °C/W ✓ (4)
$\theta_{sa} = 12.22 - 1.0 - 0.5$ ✓ $= 10.72$ °C/W (or less) ✓

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QUESTION 2

2.1 The gate capacitance must be charged and discharged quickly with high peak currents (amps), which a pin cannot give, so switching would be slow and lossy ✓✓ (4)
A high-side MOSFET's source floats at the load voltage, so its gate must be driven above the supply rail ✓; the driver uses a bootstrap circuit or an isolated supply for this ✓

2.2 $I = Q/t = 60 \times 10^{-9} / 100 \times 10^{-9}$ ✓ $= 0.6$ A ✓ (2)

2.3 Snubber: limits the rate of rise of voltage (dv/dt) and the voltage spikes across a switch when it turns off, protecting it and reducing interference ✓✓ (4)
Freewheeling diode: gives the inductive load current a path when the switch opens ✓, so the stored energy does not cause a destructive voltage spike ✓

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QUESTION 3

3.1.1 $V_m = 230\sqrt{2} = 325.27$ V ✓ (4)
$V_{dc} = (V_m / 2\pi)(1 + \cos \alpha)$ ✓ $= (325.27 / 2\pi)(1 + 0.5)$ ✓ $= 77.65$ V ✓

3.1.2 Output zero from 0° to 60°, then follows the positive half of the sine wave from 60° to 180° ✓✓; zero for the whole negative half-cycle ✓ (4)
Firing angle: the angle after the start of the positive half-cycle (the natural turn-on point) at which the SCR is triggered ✓

3.2.1 $V_{dc} = (2V_m / \pi) \cos \alpha$ ✓ $= (2 \times 325.27 / \pi) \times \cos 30°$ ✓ $= 207.07 \times 0.866$ ✓ $= 179.33$ V ✓ (5)
$I_{dc} = 179.33 / 10 = 17.93$ A ✓

3.2.2 V_{dc} becomes negative: the converter works as an inverter, sending power from the DC side back to the AC supply ✓; used for regenerative braking of DC motor drives or in HVDC links ✓ (2)

3.3 $V_{dc} = (3\sqrt{2} / \pi) V_L = 1.35 \times 400$ ✓✓ $= 540.19$ V ✓ (4)
Ripple frequency $= 6 \times 50 = 300$ Hz ✓

3.4 $V_{dc} = 1.35 V_L \cos \alpha$ ✓ $= 1.35 \times 400 \times \cos 45°$ ✓ $= 381.97$ V ✓ (3)

3.5 The capacitor only charges near the peaks of the sine wave, so the diodes conduct for short times ✓ and the mains current is drawn in narrow, high pulses ✓. This current is rich in harmonics, giving a low power factor and distortion of the supply ✓ (3)

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QUESTION 4

4.1.1 $D = V_{out} / V_{in} = 12/24$ ✓ $= 0.5$ ✓ (2)

4.1.2 $\Delta I = (V_{in} - V_{out}) D / (f L)$ ✓ $= 12 \times 0.5 / (50\,000 \times 100 \times 10^{-6})$ ✓ $= 1.2$ A ✓ (3)

4.1.3 $I_{load} = 12/6 = 2$ A ✓ (3)
$I_{max} = 2 + 0.6 = 2.6$ A; $I_{min} = 2 - 0.6 = 1.4$ A ✓
$I_{min} > 0$, so the inductor current never falls to zero: continuous conduction ✓

4.1.4 $\Delta V = \Delta I / (8 f C)$ ✓ $= 1.2 / (8 \times 50\,000 \times 47 \times 10^{-6})$ ✓ $= 0.06$ V (63.83 mV) ✓ (3)

4.2 CCM: the inductor current stays above zero for the whole switching cycle ✓ (3)
DCM: the inductor current falls to zero and stays at zero for part of each cycle, which happens at light loads or with a small inductor ✓
In DCM the output voltage no longer depends only on D ✓

4.3 $V_{out} = V_{in} / (1 - D)$, so $D = 1 - 12/36$ ✓ $= 0.67$ ✓ (4)
$P_{in} = P_{out}$: $I_{in} = 36 \times 1 / 12$ ✓ $= 3$ A ✓

4.4 $V_{out} = -V_{in} D / (1 - D) = -12 \times 0.6 / 0.4$ ✓ $= -18$ V (the polarity is reversed) ✓ (2)

4.5 $P_{loss} = (24 - 12) \times 2$ ✓ $= 24$ W ✓ (4)
$\eta = 24 / 48 = 50\%$ ✓
A switch-mode converter is often 85 to 95% efficient, so the battery lasts longer and less heat needs to be removed ✓

4.6 Drawing: four switches in an H with the motor across the middle, a freewheeling diode across each switch ✓✓ (6)
Forward: S1 and S4 on; reverse: S2 and S3 on ✓
Speed: PWM on one pair; the average voltage sets the speed ✓
Shoot-through: both switches in one leg on at once, which short-circuits the supply ✓; prevented by never switching both together and adding dead time in the driver ✓

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QUESTION 5

5.1 Square wave of ±48 V: V_rms = 48 V ✓ (3)
V_1(rms) = (4 V_dc / π) / √2 = 0.9 × 48 ✓ = 43.22 V ✓

5.2 A sine reference is compared with a high-frequency triangle wave; the switches change state where they cross, so the pulse widths vary sinusoidally ✓ (3)
Peak fundamental = m_a × V_dc ✓ = 0.8 × 400 = 320 V ✓

5.3 Rectifier (AC to DC) → DC link with capacitor (smooths the DC) → PWM inverter (DC to variable-frequency, variable-voltage AC) → motor, with a controller setting V/f ✓✓✓ (3)

5.4 A short delay after one switch in a leg turns off before the other turns on ✓, because real switches take time to turn off and would otherwise both conduct (shoot-through) ✓ (2)

5.5 A square wave contains large low-order harmonics (3rd, 5th, 7th) ✓ that cause extra heating, torque ripple and noise in the motor ✓ (2)

5.6 The motor acts as a generator and sends energy back into the DC-link capacitor, but the diode rectifier cannot return it to the mains ✓ (2)
Use a braking resistor switched across the DC link (dynamic braking), or a regenerative (active) front end ✓

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TOTAL: 100

END OF MARKING GUIDELINE